Dr. Albert G. Prinn

Curriculum Vitae

Relevant work experience

10/2025–present **Research Fellow**, *Friedrich-Alexander-Universität Erlangen-Nürnberg*, Germany.
- Computational acoustics and room acoustics

01/2022–present **Lecturer**, *International Audio Laboratories Erlangen (A joint institution of the Friedrich-Alexander-Universität (FAU) and Fraunhofer IIS)*, Germany.
- I teach a master's level course on Virtual Acoustics
- Preparation and delivery of course materials
- Supervision of student projects

05/2021–present **Senior Scientist**, *Fraunhofer IIS*, Erlangen, Germany.
- Conduct research in room acoustics and computational acoustics
- Supervision of younger researchers; technology scouting; publication of findings
- Two audio-related patents awarded

11/2017 – 04/2021 **Research Scientist**, *Fraunhofer IIS*, Erlangen, Germany.
- Room acoustics research, and development of acoustic modelling methods
- Writing of papers
- Leading younger researchers; helping manager make informed decisions

10/2016 – 10/2017 **Application Engineer**, *Siemens Industry Software*, Leuven, Belgium.
- Subject matter expert for NX Nastran (acoustical, structural, and thermal problems)
- Ensured industrial work-flows were captured and realized in the product
- Provided a link between product management and product development

12/2013–12/2015 **Research Fellow**, *University of Southampton*, Southampton, United Kingdom.
- Researched, developed, and validated computational models for duct acoustics
- Published technical reports, disseminated study outcomes at international conferences
- Teaching assistant for a 2nd year fluid mechanics course.

09/2009–11/2013 **Research Engineer**, *University of Southampton / LMS International*, Southampton, UK / Leuven, Belgium.
- Conducted research into computational methods for aeroacoustics, towards doctorate
- Developed an Adaptive-Order Finite Element Method (FEMAO) for acoustic simulations
- Devised a novel finite element vortex sheet model based on potential flow theory

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Relevant education

2009–2014 **Doctor of Engineering**, *University of Southampton*, United Kingdom.
Computational Acoustics

Thesis: Efficient Finite Element Methods for Aircraft Engine Noise Prediction
Supervisor: Dr. Gwénaël Gabard
Description: researched and developed a finite element method that uses adaptive high-order shape functions, for efficient aeroacoustic simulations. Researched and developed a novel finite element vortex sheet model, based on potential flow theory.

2008–2009 **Master of Science**, *University of Southampton*, United Kingdom.
Engineering Acoustics

Dissertation: experimental investigation of a parametric array in air; carried out to determine the feasibility of using this ultrasonic device for *in situ* sound absorption coefficient measurements.
Courses: advanced measurement techniques, analytical & numerical acoustics, electroacoustics, fundamentals of acoustics, fundamentals of aeroacoustics, fundamentals of vibration, human response to sound & vibration, noise control, signal processing, underwater acoustics.

2005–2008 **Bachelor of Science**, *Anglia Ruskin University*, United Kingdom.
Audio and Music Technology

Dissertation: acoustic measurement and characterization of a space used for live music performances; room impulse response measurements were performed and analysed, and subjective investigations were conducted
Selected courses: advanced acoustics and psycho-acoustics, analogue and digital electronics, analytical techniques, digital signal processing, synthesizer technology and techniques

2002-2003 **City and Guilds, Electronic Servicing**, *FAS Training Centre*, Cork, Ireland.
Theory and practice of analogue and digital electronics

1994–1998 **Matriculation**, *Pretoria Boys High School*, Pretoria, South Africa.
English, Afrikaans, Mathematics, Accounting, Biology, Physical science

References

Prof. Gwénaël Gabard,
Laboratoire d'Acoustique de l'Université du Mans, Le Mans, France.

Prof. Emanuël A. P. Habets,
Friedrich-Alexander Universität Erlangen-Nürnberg, Erlangen, Germany.

List of publications

Journal articles

Acoustic eigenvalue estimation using Rayleigh quotient iteration with a sparse matrix pencil. **A. G. Prinn**, E. A. P. Habets. Applied Acoustics. Volume 232, 110573, 2025.

Validation of an eigenvalue-based inverse method for estimating locally-reacting surface impedance. **A. G. Prinn**, P. Panter, A Walther, E. A. P. Habets. Applied Acoustics. Volume 228, 110332, 2025.

A study of the spatial non-uniformity of reverberation time at low frequencies. **A. G. Prinn**, C. Tuna, A Walther, E. A. P. Habets. Applied Acoustics. Volume 227, 110220, 2025.

Simulating room transfer functions between transducers mounted on audio devices using a modified image source method. Z. Xu, A. Herzog, A. Lodermeyer, E. A. P. Habets, **A. G. Prinn**. Journal of the Acoustical Society of America. 155 (1), 343-357, 2024.

A Review of Finite Element Methods for Room Acoustics. **A. G. Prinn**. Acoustics, 5(2), 367–395, 2023.

Data-driven local average room transfer function estimation for multi-point equalization. C. Tuna, A. Zevering, **A. G. Prinn**, P. Götz, A. Walther, and E. A. P. Habets. Journal of the Acoustical Society of America, 152 (6), 3635–3647, 2022.

Acoustic reciprocity in the spherical harmonic domain: A formulation for directional sources and receivers. Z. Xu, A. Herzog, A. Lodermeyer, E. A. P. Habets, **A. G. Prinn**. JASA Express Lett. 2 (12), 124801, 2022.

Estimation of Locally Reacting Surface Impedance at Modal Frequencies using an Eigenvalue Approximation Technique. **A. G. Prinn**, A. Walther, E. A. P. Habets. Journal of the Acoustical Society of America, 150, 2921–2935, 2021.

On Computing Impulse Responses from Frequency-Domain Finite Element Solutions. **A. G. Prinn**. Journal of Theoretical and Computational Acoustics, 29 (01), 2050024, 2021.

Adaptive, High-Order Finite-Element Method for Convected Acoustics. G. Gabard, H. Bériot, **A. G. Prinn**, K. Kucukcoskun. AIAA Journal, 56 (8), 1-13, 2018.

Efficient Implementation of High-Order Finite Elements for Helmholtz Problems. H. Bériot, **A. G. Prinn**, G. Gabard. International Journal for Numerical Methods in Engineering, January 2016.

Conference papers

On the Accuracy of the Image Source Method in the Presence of Two Adjacent Impedance Walls. Z. Xu, E. A. P. Habets, and **A. G. Prinn**. Forum Acusticum, Malaga, Spain, 23–26 June 2025.

Diffusion equation-based estimation of spatially non-uniform surface absorption coefficients. **A. G. Prinn** and E. A. P. Habets. Forum Acusticum, Malaga, Spain, 23–26 June 2025.

Estimating the impedance of a material sample at low frequencies. **A. G. Prinn**, R. Maestro, M. L. López-Ibáñez, M. Larrosa Navarro, and E. Latorre Iglesias. Forum Acusticum, Malaga, Spain, 23–26 June 2025.

Measures of validation for computational room acoustics. **A. G. Prinn**, Z. Xu, and E. A. P. Habets. Forum Acusticum, Malaga, Spain, 23–26 June 2025.

Simulating Sound Fields in Rooms with Arbitrary Geometries Using the Diffraction-Enhanced Image Source Method. Z. Xu, E. A. P. Habets, **A. G. Prinn**. IWAENC 2024, Aalborg, Denmark, September 2024.

Towards validating the diffraction-enhanced image source method. Z. Xu, K. Ersinadim, E. A. P. Habets, **A. G. Prinn**. Inter-Noise 2024, Nantes, France, August 2024.

Towards estimating acoustic surface impedances at low frequencies in a reverberation chamber. **A. G. Prinn**, E. Latorre Iglesias, M. López-Ibáñez, M. Larrosa Navarro, E. A. P. Habets. Inter-Noise 2024, Nantes, France, August 2024.

Estimating frequency-dependent absorption coefficients in small rooms using a diffusion model. **A. G. Prinn**, C. Tuna, A. Walther, and E. A. P. Habets. AES Europe 2024, Madrid, Spain, 2024.

Multi-Purpose Room Impulse Response Dataset Measured on a 3D Spatial Grid doc. L. Friede, C. Tuna, F. Knauff, **A. G. Prinn**, K. Ersinadim, A. Walther. AES Europe 2024, Madrid, Spain, 2024.

Acoustic eigenvalue estimation using the matrix pencil method and Rayleigh quotient iteration. **A. G. Prinn**, C. Tuna, A. Walther, and E. A. P. Habets. Forum Acusticum, Torino, Italy, 11–15 September 2023.

Validation of an eigenvalue technique for estimating locally reacting impedances at modal frequencies. **A. G. Prinn**, A. Walther, and E. A. P. Habets. Forum Acusticum, Torino, Italy, 11–15 September 2023.

A finite element study of absorption coefficient measurement at low frequencies. **A. G. Prinn**. Inter-Noise 2023, Chiba, Greater Tokyo, Japan, 20-–23 August 2023

Visualization of room reflections using a linear loudspeaker array and a single microphone. C. Tuna, **A. G. Prinn**, F. Knauff, A. Walther, E. A. P. Habets. Audio Engineering Society Convention, 148, 2020.

An Adaptive, High-Order Finite Element Method for Aeroengine Acoustics. G. Gabard, H. Bériot, **A. G. Prinn**, K. Kucukcoskun. 22nd AIAA/CEAS Aeroacoustics Conference, June 2016, Lyon, France.

The Effect of Steady Flow Distortion on Noise Propagation in Turbofan Intakes. **A. G. Prinn**, R. Sugimoto, R. J. Astley. 22nd AIAA/CEAS Aeroacoustics Conference, June 2016, Lyon, France.

Predicting the Effect of Flow Distortion on In-duct Acoustic Propagation. **A. G. Prinn**, R. J. Astley, R. Sugimoto. 22nd International Congress on Sound and Vibration, July 2015, Florence, Italy.

The Effect of Steady Flow Distortion on Acoustic Mode Propagation in a Straight Annular Duct. **A. G. Prinn**, R. J. Astley, R. Sugimoto, Z. Rarata. FFT Acoustic Simulation Conference, October 2014, Brussels, Belgium.

On the Performance of High-order FEM for Solving Large-scale Industrial Acoustic Problems. H. Bériot, **A. G. Prinn**, G. Gabard. 20th International Congress on Sound and Vibration, July 2013, Bangkok, Thailand.

Implementation of a Vortex Sheet in a Finite Element Model Based on Potential Theory for Exhaust Noise Predictions. **A. G. Prinn**, G. Gabard, H. Bériot. 18th AIAA/CEAS Aeroacoustics Conference, June 2012, Colorado, USA.

Finite Element Simulation of Noise Radiation Through Shear Layers. **A. G. Prinn**, G. Gabard, H. Bériot. Acoustics 2012 Conference, April 2012, Nantes, France.

Thesis

Efficient Finite Element Methods for Aircraft Engine Noise Prediction. **A. G. Prinn**. Doctoral Thesis, University of Southampton, ISVR, July 2014.

Patents

Method for determining a sound field. **A. G. Prinn**, A. Walther, E. A. P. Habets, Y. El Baba. 2023.

Apparatus and method for automatic adaption of a loudspeaker to a listening environment. **A. G. Prinn**, A. Walther, C. Tuna. 2021.